OXFORD

INTERNATIONAL AQA EXAMINATIONS

Topic Test: OxfordAQA International AS level Physics

Electricity

Name:

Class:

Date:

Time: **56 minutes**

Marks: **37 marks**

Comments:

1 During a lightning strike, 9.4×10^{18} electrons move from a cloud to the ground in a time of 18 μs.

(a) Calculate the current in the lightning strike due to the transfer of electrons.

current = ________________________ A

(2)

(b) State the direction of the conventional current during this lightning strike.

____

____

(1)

(Total 3 marks)

2 (a) Define resistance.

____

____

____

(1)

(b) (i) Sketch onto the axes below a graph of the variation of current with potential difference for a filament lamp.

(1)

(ii) State and explain, in terms of electron flow, how the resistance of the filament lamp changes as the current in the lamp increases.

____

____

____

____

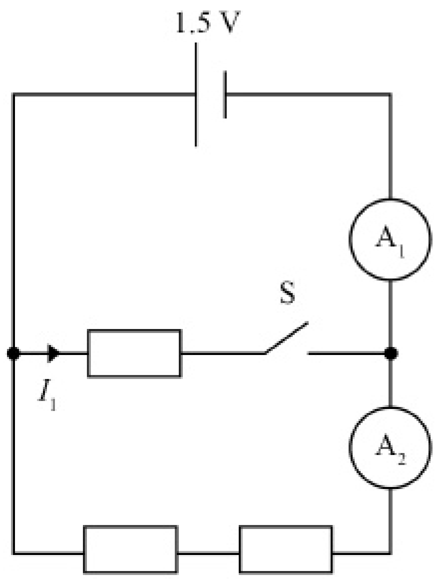
____

____

(3)

(Total 5 marks)

3 The diagram shows three identical resistors, each with a resistance R, in a circuit connected to a cell with emf 1.5 V
The cell has negligible internal resistance.



(a) When switch S is closed the reading of ammeter A_1 is 1.2 A and the reading of ammeter A_2 is 0.40 A

State the value of the current I_1 when switch S is closed.

I_1 = ____________________ A

(1)

(b) Calculate R.

R = ____________________ Ω

(1)

(c) Switch S is returned to the open position.

State and explain the effect this has on the readings of A_1 and A_2.

____

____

____

____

____

____

____

____

(3)

(Total 5 marks)

4 **Figure 1** shows a circuit used to determine the emf and internal resistance of a cell.

Figure 1

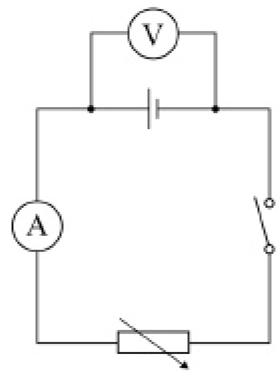
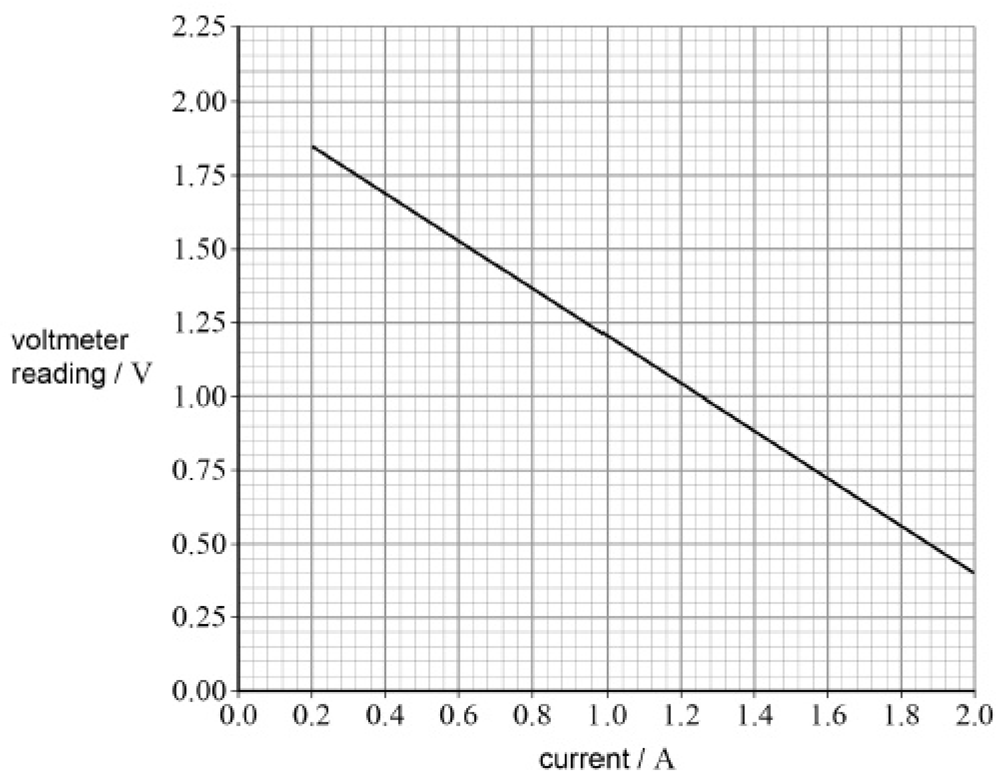


Figure 2 shows the variation of the voltmeter reading with current in the circuit as the variable resistor is adjusted.

Figure 2



(a) The circuit contains an ideal voltmeter and an ideal ammeter.

State the resistance of an ideal voltmeter.

____

____

(1)

(b) Show that the internal resistance of the cell is approximately 0.8 Ω.

(1)

(c) The variable resistor is adjusted until the current in the circuit is 2.10 A.

Calculate the resistance of the variable resistor.

resistance = ____________________ Ω

(3)

(Total 5 marks)

5 A fuse wire is a short piece of wire in series with a component in a circuit. The fuse wire is designed to melt and act as a circuit breaker when the current exceeds a safe level. A lighting technician needs to choose a fuse wire for a spotlight that normally operates at 11 A. He can choose from two types of fuse wire, **A** and **B**, shown in the table below. Each fuse wire is 1.2 cm long.

Fuse wire	**Metal**	**Resistivity / ....**	**Cross-sectional area / m^2**	**Power required to melt fuse wire / W**
A	Zinc	5.5×10^{-8}	1.8×10^{-8}	2.4
B	Silver	1.6×10^{-8}	9.5×10^{-9}	2.7

(a) Complete the table above with the SI unit of resistivity.

(1)

(b) Determine which fuse wire should be used for the spotlight.

fuse wire = ______________________

(4)

(c) Suggest **one** other factor that might influence the technician's choice.

____

____

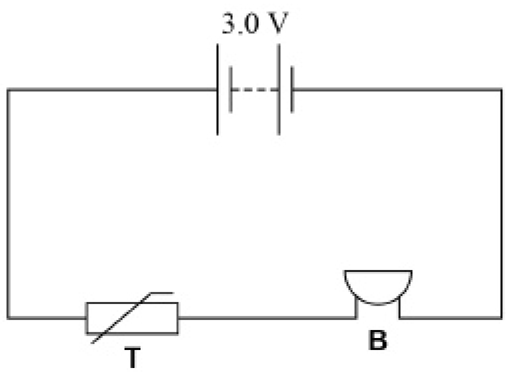
(1)

(Total 6 marks)

6 **Figure 1** shows a thermistor **T** used in an alarm circuit for a refrigerator. The alarm is designed to sound a buzzer **B** when the temperature exceeds a threshold value. **B** has a constant resistance of 123 Ω.

The battery has a negligible internal resistance.

Figure 1



The buzzer sounds when the potential difference across it is greater than 1.8 V.

(a) Explain why the potential difference across the buzzer increases when the temperature increases.

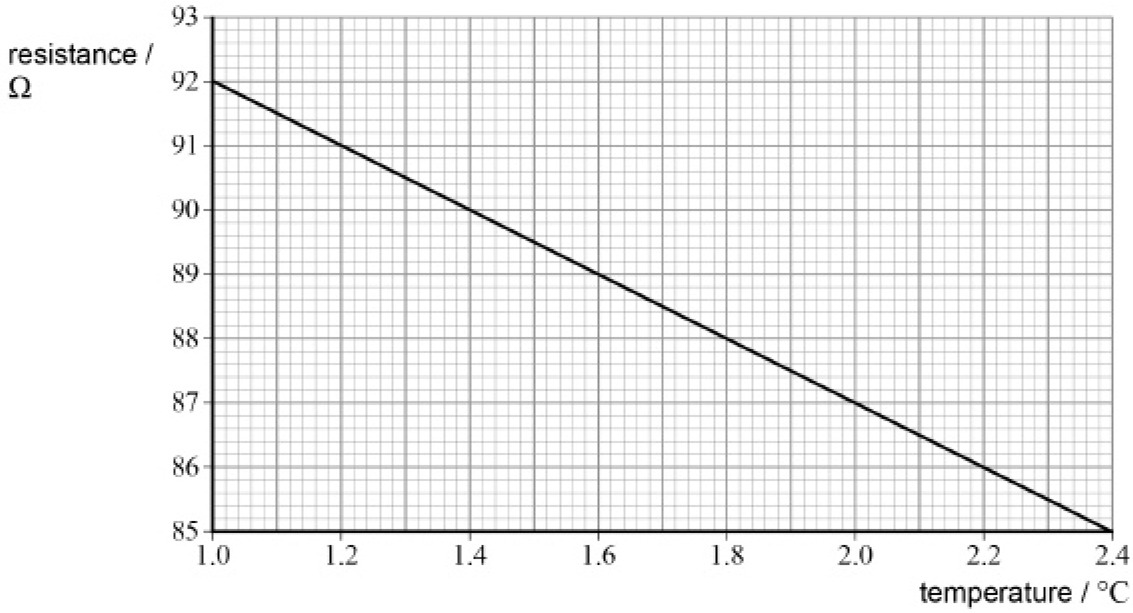
(3)

(b) Show that, when the potential difference across the buzzer is 1.8 V, the resistance of the thermistor is about 80 Ω.

(2)

Figure 2 shows how the resistance of the thermistor varies with temperature over a small range of temperatures.

Figure 2



(c) Calculate the gradient of the graph.

gradient = ________________________ $\Omega\ K^{-1}$

(2)

(d) Determine the temperature at which the buzzer will start to sound.
Assume that the gradient of the graph remains constant when the temperature rises above 2.4 °C.

temperature = ________________________ °C

(1)

(Total 8 marks)

7 What is a unit for potential difference?

A A Ω^{-1} ⬭

B C J^{-1} ⬭

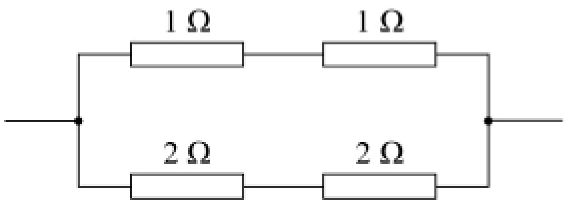
C J A^{-1} s^{-1} ⬭

D W A ⬭

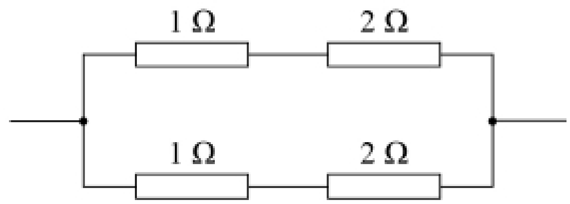
(Total 1 mark)

8 Which network of resistors has the largest total resistance?

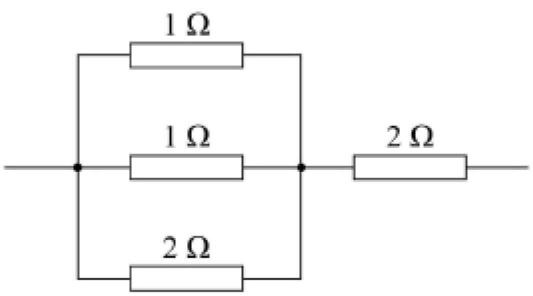
A



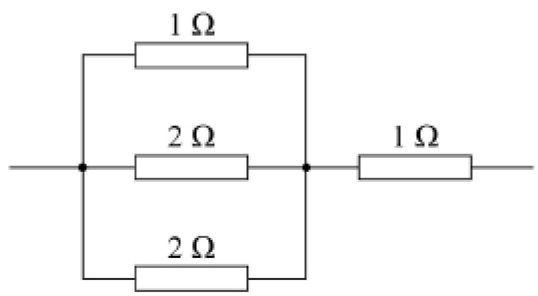
B



C



D



A ☐

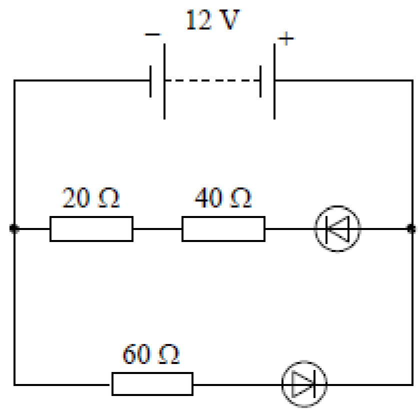
B ☐

C ☐

D ☐

(Total 1 mark)

9 The 12 V battery in the circuit shown has negligible internal resistance. The diodes have 'ideal' characteristics.



The current through the battery is approximately

A 0 A

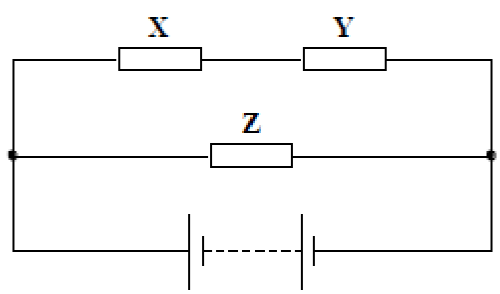
B 0.10 A

C 0.20 A

D 0.40 A

(Total 1 mark)

10 Three identical resistors **X**, **Y** and **Z** are connected across a battery as shown.



The ratio $\dfrac{\text{power developed in } \mathbf{X}}{\text{power developed in } \mathbf{Z}}$ is

A $\frac{1}{4}$

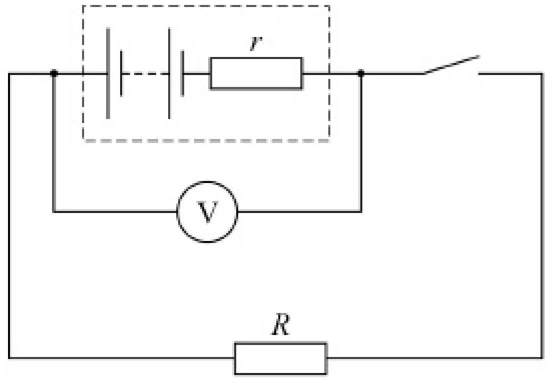
B $\frac{1}{2}$

C 1

D 2

(Total 1 mark)

11 The diagram shows a 12 V battery connected to a resistor of resistance R.
The voltmeter reads 10 V when the switch is closed.



What is the internal resistance r of the battery?

A $\frac{R}{6}$ ○

B $\frac{R}{5}$ ○

C $5R$ ○

D $6R$ ○

(Total 1 marks)

Mark schemes

1 (a) I = Q/t = $9.4 \times 10^{18} \times 1.6 \times 10^{-19} / 18 \times 10^{-6}$ ✓

= 84 000 A (83 556 A) ✓

2

(b) Up/upwards/from ground to cloud ✓

1

[3]

2 (a) ratio of voltage (across component) to current (through component) or R = V/I **with** terms defined and R as subject

B1

1

(b) (i) correct curve

B1

1

(ii) resistance increases / increase in resistivity

B1

energy **transfer increases lattice vibration/** temperature rise **increases lattice vibration** / electron collisions **increases lattice vibration**

B1

more **frequent** collisions/ ions now a larger target for electrons

B1

3

[5]

3 (a) I_1 = 0.80 A

Accept 1 sig. fig. in this case

1

(b) R = 1.9 Ω

1

(c) effect : Reading of A_1 decreases

1

reason: because the total circuit resistance has increased

(reason marks are dependent on effect marks)

1

effect: Reading of A_2 remains constant

1

reason: The p.d. and resistance for this branch are constant.

Accept lower part of circuit is unaffected by the switch

1

Max 3

Alternative:

$A_1 = A_2$

1

Total circuit resistance = $2R$

1

Current = 0.40 A

1

So A_1 decreases (A_2 stays the same)

1

Max 3

[6]

4 (a) infinite ✓

Condone "very large"

1

(b) Evidence of using graph to get gradient

OR

use of intercept (2 V) and a data point e.g (1, 1.2) ✓

Must be a correct pair of coordinates.

1

(c) E = 2.00 (V) ✓

Substitution into $E = I(R + r)$ (e.g candidate's $E = 2.1(R + 0.8)$) ✓

0.15(2) (Ω) ✓

OR

Total R = 2.00/2.1 = 0.95 ✓

R = 0.95 − 0.8 ✓

0.15(2) (Ω) ✓

Alternative method:

Deduce pd correctly (0.32 V) and substitute into $R = V/I$ = 0.32/2.1

Allow reasonable range for data extraction

Allow 1 max for any extrapolation done by extending the grid and line but reward a "mathematical" extrapolation such as using similar triangles.

3

[5]

5 (a) Ωm

1

(b) Use of $R = \rho l/A$ seen

Allow reverse argument i.e. calculating power needed to melt fuses starting from 11 A

Correct value for at least 1 resistance (zinc 0.037 Ω, silver 0.020 Ω) ✓

Use of $P = I^2R$ with a correct substitution of their R ✓

Correct value for both currents (8.1 A for zinc 11.6 A for silver) or powers and statement justifying the use of silver ✓

4

(c) Any sensible factor eg corrosion, cost, availability

1

[6]

6 (a) As temperature increases, the resistance of the thermistor decreases ✓

As more charge carriers become available for conduction/ are raised to the conduction band ✓

A larger share/proportion of the battery pd across the buzzer (or reference to a potential divider) ✓

3

(b) Use of $\frac{1.2}{1.8} = \frac{R}{123}$ or $I = \frac{1.8}{123}$ or I = 0.0146 (A) ✓

Must see 82 not just 80. Answers based on checking can only score both marks if unrounded values are seen.

R = 82 (Ω) ✓

2

(c) Attempt to find gradient as Δy/Δx with correct values read from graph

Gradient = −5 ($\Omega\ K^{-1}$)

Must have negative sign for second mark.

2

(d) T = 3.0 (°C)

1

[8]

7 C

[1]

8 C

[1]

9 C

[1]

10 A

[1]

11 B

[1]